

GERTIAN MICI

Senior Backend Engineer — .NET, Distributed Systems, Payments & AI Engineering

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Swedish work permit | English (C1, IELTS) | Albanian (native)

PROFESSIONAL SUMMARY

Senior backend engineer with 7 years building high-availability, cloud-native services in .NET, now working at both ends of AI engineering. Deep experience in consumer fintech — card issuing, ledgers, settlement and payments — across microservices, event-driven architecture and a hand-written data access layer with no ORM, where the schema, indexes, stored procedures and versioned migrations are mine. Technical lead for a consumer debit card portfolio migration between issuing processors, covering the migration pipeline, the cutover safety design and the operational runbooks. Equally focused on production LLM systems and the tooling that builds software: assistant tool layers, LLM observability and cost reporting, and autonomous developer agents. I verify before I claim, and I fix the whole class of a bug rather than the single instance reported.

TECHNICAL SKILLS

Languages: C#, T-SQL, Python, TypeScript, JavaScript, Swift

Backend & Architecture: .NET 8/7/6/Core, ASP.NET Core Web API, microservices, event-driven architecture, CQRS, clean architecture, REST and gRPC APIs (Protobuf), SignalR, domain-driven design

AI & LLM Engineering: Agent tool design, agent orchestration, LLM observability and cost reporting, prompt-injection defence, structured model output (JSON Schema), on-device inference (MLX), AI development harnesses, Claude Code, Cursor, GitHub Copilot, Kiro

Distributed Systems: RabbitMQ, Kafka, transactional outbox, idempotency and retry design, distributed locking, retryable-vs-permanent failure handling, rate-limit handling, distributed tracing

Data: SQL Server, Azure SQL, PostgreSQL, MongoDB, Redis, schema and index design, stored procedures, versioned migrations, query and index optimisation, BigQuery analytics SQL

Cloud & DevOps: Azure (SQL Database, Blob Storage, Key Vault, AKS), Docker, Kubernetes, Argo CD / GitOps, GitHub Actions

Security: PCI-scoped development, applied cryptography, key management and rotation, OAuth2, IdentityServer4 (PKCE), secrets handling, least-privilege data access, PII masking

Observability & Testing: Elasticsearch, Kibana, correlation-id tracing, structured logging, MSTest, xUnit, FluentAssertions, integration and cross-service regression testing, BenchmarkDotNet

PROFESSIONAL EXPERIENCE

Senior Software Engineer, MAJORITY

Stockholm, Sweden | Jan 2024 – Present | Consumer fintech: card issuing, ledgers, settlement and payments

- **Card processor migration, technical lead.** Moved a consumer debit card portfolio between card-issuing processors, one of the higher-risk changes a card programme can attempt. Covered the restartable migration pipeline, the safety design for the cutover window so customers kept service throughout, staged and reversible rollout, and operational runbooks written and circulated in advance.
- **Settlement and ledger correctness.** Investigating how transactions are authorized, cleared, settled and posted, resolving discrepancies found in production, and closing each one with MSTest regression tests so it cannot reappear.
- **In-app AI assistant, money and transaction layer.** Safe access to transaction and statement data through 5 agent tools. Access is limited to the signed-in customer by construction, because the identity comes from their token rather than from anything the model can supply, and the read path was kept off the production transactions database.
- **AI development tooling.** A reusable harness that enforces verifying work before calling it finished and finding the root cause before fixing, with 8 focused review agents and 11 workflow commands; and an agent that takes a ticket through to an opened pull request unattended across 3 platforms, under a spend limit and in an isolated working copy. Adopted by another platform team.
- **Failure classification across services.** Separated permanent failures from retryable ones so callers stop retrying requests that can never succeed, and rolled it out without changing the behaviour of any existing handler on deploy. Own the messaging underneath: RabbitMQ handlers, the transactional outbox, and idempotency keys on retryable work.
- **New credit service and crypto-backed wallet.** One of the first engineers on a consumer credit service, from an empty repository to live account creation and card issuing within 2 quarters. Backend for a crypto-backed global wallet with a staged rollout behind feature flags, including Solana USDC transactions via Privy wallet.
- **Cardholder-data cryptography, PCI scope.** Card fingerprinting, sensitive credentials designed so they are never stored, and key rotation carried through to retiring the previous scheme. Consolidated duplicated hashing logic into a shared platform library and migrated 7 downstream services onto it in one coordinated release.
- **Production incident investigation.** Daily in Elasticsearch and Kibana — error triage, correlation-id traces and call-tree reconstruction across service hops — and packaged that workflow into a reusable AI skill, so an agent now runs the first pass of an investigation instead of it being done by hand.

- **Developer tooling and platform.** LLM observability and cost reporting, plus platform work: a .NET major-version upgrade across 6 backend services in 2 weeks and a move to Kubernetes-native GitOps delivery, each service shipping as a Docker container onto AKS, over Azure SQL Database, Blob Storage and Key Vault.

Senior Software Engineer (Contract), Qbrick

Stockholm, Sweden | Sep 2023 – Jan 2024

- Stabilised event-driven services: resolved memory allocation issues and eliminated throughput bottlenecks under sustained load.
- Reduced database read latency by roughly 50% through index and query optimisation.
- Strengthened CI/CD pipelines and health checks to improve release confidence.

Software Engineer / Team Lead, Lufthansa Industry Solutions

Tirana, Albania | Jun 2022 – Sep 2023

- Led backend development for a large-scale navigation platform covering routing, rewards, favourites and vehicle domains.
- Built .NET microservices communicating over gRPC and Kafka, with Redis, PostgreSQL and MongoDB for persistence and caching, and xUnit suites covering the service layer.
- Supported onboarding of 70,000+ users within 48 hours with no customer-impacting incidents; defined reliability targets and the dashboards to track them.

Software Developer, S2 Global (an OSI Systems company)

Tirana, Albania | Mar 2021 – Jun 2022

- Developed high-availability cloud services and implemented single sign-on with IdentityServer4 (Authorization Code with PKCE).
- Worked alongside an Angular team on UI integration requirements and authentication flows.

Full-stack Software Developer, Soft & Solution

Tirana, Albania | Sep 2019 – Feb 2021

- Delivered .NET MVC and SignalR applications plus a Xamarin mobile application across small, fast-moving teams.
- Built real-time features and maintained SQL Server performance for transactional workloads.

SELECTED PROJECTS

- **Everyday — LLM language-immersion game (iOS).** Swift, SwiftUI, structured model output. Every character is a language model held to the learner's CEFR level, with a goal to reach in the target language. Bundled content stores the intent of an exchange rather than any target-language text, so adding a language is configuration rather than a translation project — enforced by a test. The whole turn (line, translation, emotion, goal state, new vocabulary) returns from one call against a strict JSON schema, so decoding has no optionals and no retry path.
- **Helkost — offline food-processing scanner for iOS (gertianmici.com/helkost).** Swift, SwiftUI, SwiftData, CloudKit, HealthKit. Ships 1.1M products inside the app, so a barcode resolves with no network. The build pipeline cuts a 1.2 GB export to the rows carrying a processing verdict and writes them to a WITHOUT ROWID table in primary-key order, so the table is its own index and no page splits happen. Barcodes outside it fall back to the live API.
- **Osrever — iOS reverse-audio party game (gertianmici.com/osrever).** SwiftUI, AVFoundation. One player records a word; the app reverses the PCM samples for a second player to mimic, then reverses their attempt to reveal how close they got. Entirely on-device: no networking, no analytics, no third-party SDKs, and a privacy manifest declaring zero data collection. Ships via a GitHub Actions pipeline that signs with an App Store Connect API key and promotes a release branch to TestFlight.
- **ClipClap — macOS clipboard manager (github.com/GertianMici/ClipClap).** Swift, AppKit. Windows-style clipboard history for macOS: recall with a global hotkey, search the history, and paste by number key. Open source.
- **Browser games — nine games, no engine (gertianmici.com/#/games).** Three.js and Canvas. 3D racing and chess, a maze arcade game, a cellular-automaton sand sandbox, a one-button stacker and two idle games, each written from scratch with sound synthesised at runtime.

EDUCATION & CERTIFICATIONS

BSc, Computer Engineering, Polytechnic University of Tirana

Tirana, Albania | Oct 2017 – Aug 2020

Microsoft Certified: Azure Fundamentals (2023) · Microsoft Exam 480: HTML5 with JavaScript and CSS3 (2020)